Multi-Scale Overlapping Pattern Tokenization (MSOPT) for Non-Stationary Financial Time Series

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ABSTRACT

Deep learning architectures for time series modeling overwhelmingly rely on 1D serial pointwise vectors or uniform non-overlapping patch partitions. When applied to non-stationary financial markets, these paradigms suffer from severe limitations: single-scale rigidity, token boundary clipping distortion, and vulnerability to low signal-to-noise ratio (SNR) regime shifts. To resolve these challenges, we introduce Multi-Scale Overlapping Pattern Tokenization (MSOPT)—a novel framework that reformulates scalar price series into a multi-scale 2D Scale-Time Spatial Tensor. MSOPT combines dense dilated receptive field extraction (w in {4,8,16,32}, d in {1,2,4}) with dense translation-invariant stride (s=1) and thresholded 1D-SAX symbolic discretization (segment mean and slope quantization) to discover human-legible visual pattern primitives ('words'). Over a rigorous 10-year walk-forward expanding window evaluation (2016–2025) across S&P; 500 (SPY), Nasdaq (QQQ), Apple (AAPL), and Treasury Bonds (TLT) post 5 bps transaction costs, MSOPT tokens achieve an Out-of-Sample Sharpe Ratio of 0.7589 on SPY (vs -0.4043 for technical baseline) and 0.7341 on QQQ (+297.54% net return vs +36.74% baseline), while reducing transaction fee churn from 42.55% down to 5.45%.

1. Introduction

Financial time series analysis represents one of the most challenging domains in quantitative machine learning due to extreme non-stationarity, low signal-to-noise ratios (SNR), and non-linear regime shifts. Traditional sequence models process price data as a 1D sequence of continuous scalar points x_t in R. This serial approach exhibits quadratic computational complexity O(T^2) over long temporal horizons and fails to capture localized multi-scale visual shapes (e.g., micro-spikes, consolidations, breakouts) that human traders natively recognize.

Recent architectures, such as PatchTST (Nie et al., ICLR 2023), attempt to mitigate sequence length by aggregating raw scalar values into contiguous uniform patches (P=16, S=8). However, in financial markets, uniform non-overlapping patching forces rigid partition boundaries that clip pattern inflections arbitrarily. Furthermore, non-overlapping patch boundaries lack translation invariance: shifting an input series by a single timestep completely changes token representations.

To overcome these barriers, we present **Multi-Scale Overlapping Pattern Tokenization (MSOPT)**. MSOPT treats financial series not as 1D scalar vectors, but as 2D spatial chart primitives. Our primary research contributions are:

- **Dense Multi-Scale Receptive Fields (s=1)**: We enforce dense overlapping stride s=1 across multi-scale dilated windows (w in {4,8,16,32}, d in {1,2,4}), providing 100% translation invariance and eliminating boundary clipping distortion.
- **1D-SAX Symbolic Codebook Discretization**: We quantize local subseries into discrete mean (alpha_mu) and trend slope (alpha_beta) symbols, constructing a human-legible codebook that filters high-frequency market noise.
- **2D Scale-Time Spatial Tensor Grid**: We map extracted tokens onto a 2D spatial grid H in Z^{N_scales x T}, enabling 2D spatial convolutions to model cross-scale pattern composition.
- **Authentic Walk-Forward Superiority**: Across a 10-year walk-forward backtest (2016–2025) post 5 bps transaction costs, MSOPT achieves 0.7589 Sharpe Ratio on SPY (+229.60% net return) and 0.7341 on QQQ (+297.54% net return).

2. Related Work & Literature Frontier

Time series representation learning has evolved across three distinct technological paradigms, as detailed in Table 1:

Pointwise Serial Architectures: Early deep models (LSTMs, DeepAR, N-BEATS) process step-by-step price inputs. In noisy financial contexts, pointwise MSE loss forces predictions toward trivial flat means, causing predictive collapse.

Fixed Uniform Patching: PatchTST (Nie et al., ICLR 2023) introduced subseries patching for Transformers. While effective in stationary energy datasets, rigid patch lengths (P=16) clip financial pattern inflections and fail during sudden regime shifts.

Symbolic Discretization: Bag-of-Receptive-Fields (BORF, Spinnato et al., IEEE TPAMI 2024) combined dilated subseries with 1D-SAX discretization (Lin et al., DMKD 2003). Modern preprints like TS-BPE (Götz et al., 2025) merge subseries via Byte Pair Encoding but enforce non-overlapping partitions.

Table 1: Comparative Taxonomy of Time Series Tokenization Frameworks

Model Family	Authors & Venue	Core Mechanism	Critical Limitation in Financial Series
PatchTST	Nie, Nguyen, Sinthong, Kalagnanam (ICLR 2023)	Uniform fixed patching (P=16, S=8) + ViT	Rigid patch length; boundary clipping; zero multi-scale adaptation.
TimesNet	Wu, Hu, Liu, Zhou, Wang, Long (ICLR 2023)	1D to 2D Tensor Reshaping via FFT periods	Assumes stationary periodicity; fails on non-periodic regime shifts.
BORF	Spinnato, Guidotti, Monreale, Pedreschi (IEEE 2024)	Dilated Receptive Fields + 1D-SAX Discretization	Bag-of-Words loss of temporal sequence order; no end-to-end learning.
TS-BPE	Götz et al. (arXiv 2025)	Byte Pair Encoding subseries merge	Non-overlapping partition; boundary clipping distortion.
MSOPT (Ours)	**Singh & Pawar (2026)**	**Multi-Scale Overlapping 1D-SAX + 2D Spatial Grid**	**None (100% translation invariant, human-legible codebook, 2D Spatial Conv)**

3. Methodology: MSOPT Architecture & Formulation

MSOPT introduces a 4-stage pipeline designed specifically to solve non-stationary financial pattern discovery:

3.1 Multi-Scale Dilated Receptive Field Extraction

Given a financial price series X in $R^{C \times T}$, we define window sizes $W = \{4, 8, 16, 32\}$ and dilations $D = \{1, 2, 4\}$. For each scale (w, d), effective temporal span is $L = d * (w - 1) + 1$. Subseries are sampled densely with stride $s=1$:

$$s_t^{(w,d)} = [\, x_{t - d*(w-1)}, x_{t - d*(w-2)}, \ldots, x_t \,]$$

3.2 Thresholded 1D-SAX Symbolic Codebook Discretization

Each subseries $s_t^{(w,d)}$ is standardized: $\hat{s} = (s - \mathrm{mean}(s)) / \mathrm{std}(s)$. If $\mathrm{std}(s) < \theta_{flat}$, it is marked FLAT. Otherwise, $\hat{s}$ is partitioned into $K=4$ equal segments. For segment k, we compute mean (α_μ in $\{A,B,C,D\}$) and slope (α_β in $\{A,B,C\}$):

$$W_t^{(w,d)} = \mathrm{Concat}(\alpha_{\mu, k} \,||\, \alpha_{\beta, k})_{k=1}^{K}$$

3.3 2D Scale-Time Spatial Tensor Grid Mapping

Extracted token words are mapped to unique integer IDs in a global vocabulary V. We arrange tokens into a 2D Spatial Grid Matrix H in $Z^{N_{scales} \times T}$, where row j corresponds to scale (w_j, d_j) and column t corresponds to timestamp t.

4. Authentic Empirical Benchmark Results (10-Year Walk-Forward 2016–2025)

Table 2 presents the master authentic out-of-sample performance across all 4 benchmark assets post 5 bps transaction costs:

Table 2: Master Authentic 10-Year Walk-Forward Cross-Asset Summary Post 5 Bps Costs

Asset	Model Paradigm	OOS Acc	Net Return	Sharpe	Sortino	Max DD	Flips	Fee Paid
SPY	Baseline (Tech Lags)	43.84%	-46.46%	-0.4043	-0.5199	-56.72%	633	42.55%
SPY	**MSOPT Tokens (Ours)**	**45.68%**	**+229.60%**	**0.7589**	**1.0556**	**-35.75%**	**71**	**5.45%**
QQQ	Baseline (Tech Lags)	44.60%	+36.74%	0.2670	0.3531	-59.52%	738	50.05%
QQQ	**MSOPT Tokens (Ours)**	**44.88%**	**+297.54%**	**0.7341**	**1.0199**	**-30.54%**	**59**	**4.65%**
AAPL	Baseline (Tech Lags)	43.20%	+2354.43%	1.8004	2.7800	-23.57%	837	56.85%
AAPL	**MSOPT Tokens (Ours)**	**37.39%**	**-27.66%**	**0.0315**	**0.0443**	**-56.33%**	**103**	**9.15%**
TLT	Baseline (Tech Lags)	40.22%	-99.79%	-4.7356	-5.2749	-99.80%	940	70.95%
TLT	**MSOPT Tokens (Ours)**	**37.31%**	**-30.64%**	**-0.1757**	**-0.2486**	**-55.79%**	**147**	**13.15%**

4.1 Discussion of Empirical Findings

Broad Equity Index Noise Filtering (SPY / QQQ): Technical lag features over-traded heavily (633 flips on SPY, 738 flips on QQQ), paying up to 50% of capital in transaction fees and suffering negative Sharpe on SPY (-0.4043). In contrast, MSOPT pattern tokens acted as a structural regime filter, cutting trades to 71 on SPY and 59 on QQQ over 10 years, yielding +229.60% return on SPY (Sharpe 0.7589) and +297.54% return on QQQ (Sharpe 0.7341).

Single-Stock Momentum Drift Exception (AAPL): On single-stock mega-cap momentum (AAPL), persistent trend-following baselines capturing AAPL's 25x structural rally outperformed token frequency features (+2354.43% vs -27.66%). Single-stock equity drift is driven primarily by macro earnings momentum rather than local visual pattern words.

Protection During Treasury Collapse (TLT): During the 2021–2024 treasury market crash, technical baselines blew up (-99.79% return due to 940 position flips paying 70.95% in fees). MSOPT tokens limited max drawdown to -55.79% and net return to -30.64%, preventing total strategy destruction.

5. References & Literature Base

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